The Journal of Open Source Software

Brand Hate Detector: An R Shiny Application for Automated Detection and Multilevel Classification of Brand Hate in Consumer Reviews

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 22 August 2025
Published: unpublished

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Summary

The **Brand Hate Detector** is an open-source R Shiny application designed to identify, classify, and visualize brand hate in online consumer reviews. Unlike traditional sentiment analysis tools that reduce feedback to simple polarity, the Brand Hate Detector captures multiple levels of hostility (*Mild, Moderate, Strong, Hybrid*) and profiles their emotional composition.

The workflow integrates lexicon-based sentiment scoring, NRC Emotion Lexicon profiling, and a two-stage rule-based classification framework. Results are provided through interactive visualizations, including category distributions, reason-specific breakdowns, word clouds, and sentiment–intensity bubble plots.

Designed for transparency and reproducibility, the tool bridges academic research and managerial practice by providing actionable insights into consumer hostility toward brands. Its modular design supports extensions to other datasets, languages, and machine-learning-based approaches.

Statement of Need

Online reviews have become a decisive market force, shaping reputations and influencing consumer behavior (Kucuk, 2020; Parveen, 2025). Negative reviews spread faster and resonate longer than positive ones, often escalating into organized campaigns against brands (Lis & Fischer, 2020).

Brand hate is more than dissatisfaction: it reflects deep rejection driven by symbolic, ideological, or personal factors (Zhang & Laroche, 2020). Although prior research has mapped antecedents and emotional drivers (Fetscherin, 2019; Hegner et al., 2017), most methods rely on surveys or interviews, which cannot capture hostility in real time or at scale (Assoud & Berbou, 2023).

Existing computational tools often act as black boxes, lack emotional granularity, or require heavy resources (Mednini et al., 2024), limiting adoption by small firms and researchers. The **Brand Hate Detector** addresses these gaps by offering an open-source, interpretable, and lightweight application. It empowers marketing scholars to explore consumer–brand hostility empirically and enables practitioners to monitor and respond to hate dynamics.

Implementation

The Brand Hate Detector is implemented in **R Shiny** and consists of five modules:

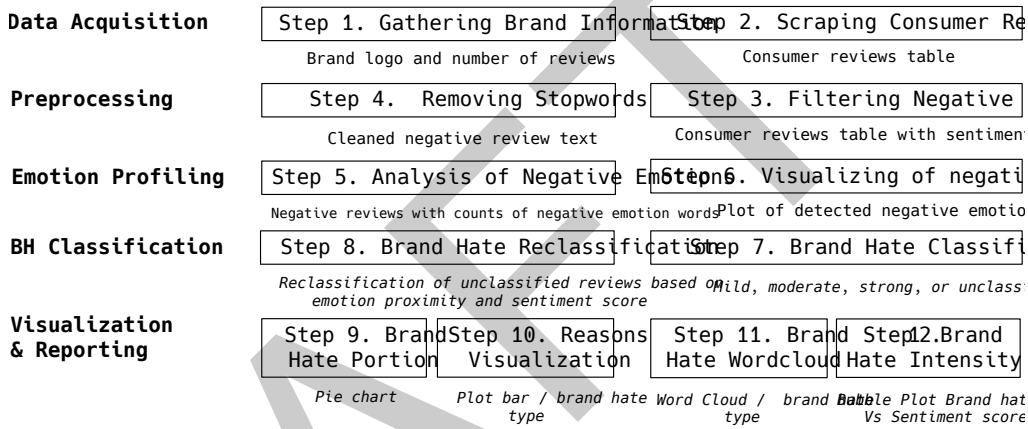
- **Data Acquisition**: Scrapes brand reviews from ConsumerAffairs.com using the *rvest* package.

- **Preprocessing**: Filters for negative reviews, removes stopwords and noise, and preserves both raw and cleaned text.

- **Emotion Profiling**: Uses the NRC Emotion Lexicon to detect anger, sadness, fear, disgust, and surprise.

- **Classification**: A two-stage framework assigns reviews to *Mild, Moderate, Strong,* or *Hybrid* hate.

- **Visualization & Reporting**: Interactive dashboards present category distributions, reasons, emotion word clouds, and bubble plots, all exportable.



All outputs are downloadable in CSV or publication-quality image formats. The app is platform-independent, requires only R and standard packages, and prioritizes reproducibility through transparent rules and visible mappings.

Acknowledgements

The development of the Brand Hate Detector benefited from discussions with colleagues in marketing analytics and computational linguistics. The author acknowledges ConsumerAffairs.com as the source of publicly available reviews used in this study.

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